

# Data Preprocessing

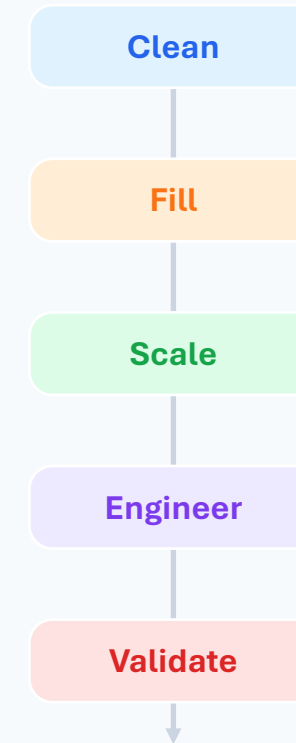
Turning messy data into model-ready features



# By the end, students should be able to...

- 1 Identify common data quality issues**  
duplicates, wrong types, outliers, inconsistent values
- 2 Choose a missing-value strategy**  
drop, impute, or add missingness indicators
- 3 Select an appropriate scaler**  
min-max, z-score, or robust scaling
- 4 Explain basic feature engineering**  
new features, encodings, interactions, polynomials

## SESSION FLOW



# The preprocessing pipeline in one picture



**Do not treat preprocessing as a one-time “cleanup script.” In production ML, it becomes part of the data and model lifecycle.**

Practical rule: the same transformations used in training must be reproducible during validation, testing, and inference.

# Data cleaning starts with asking “what is wrong here?”

1

## Duplicates

Repeated rows or entities that bias counts and training.

2

## Incorrect types

Dates as strings, numbers as text, booleans as categories.

3

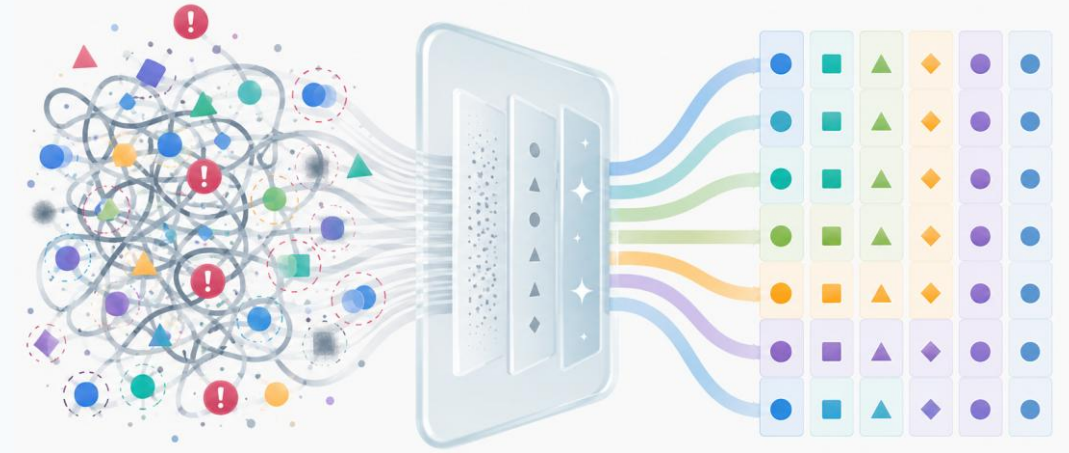
## Outliers

Extreme values: real signal, measurement error, or rare event?

4

## Inconsistencies

Different labels for the same thing: “USA”, “U.S.”, “United States”.



# Typical cleaning actions: simple, but high impact

## 1. Inspect

Profile columns, distributions, nulls, unique values

## 2. Correct

Cast types, standardize labels, fix units and formats

## 3. Deduplicate

Decide row-level vs. entity-level duplicates

## 4. Treat outliers

Validate, cap/winsorize, transform, or keep with explanation

### EXAMPLE: LABEL STANDARDIZATION

| Raw           | Clean |
|---------------|-------|
| U.S.          | USA   |
| United States | USA   |
| usa           | USA   |
| US            | USA   |

# Missing values are not all the same

## MCAR

### Missing Completely at Random

The missingness is unrelated to observed or unobserved values.

Example: random sensor upload failure.

## MAR

### Missing at Random

Missingness is related to another observed variable.

Example: income missing more often for younger users.

## MNAR

### Missing Not at Random

Missingness depends on the missing value itself or hidden causes.

Example: high earners omit income more often.

# How to handle missing values: a practical toolbox

## Drop rows/columns

Use when missingness is small, random, and removal does not bias the dataset.

## Advanced imputation

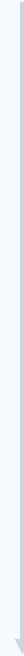
KNN, regression, or multivariate methods when patterns are richer.

## Simple imputation

Mean / median / mode. Fast baseline; median is safer with skew/outliers.

## Missingness indicator

Add a binary flag when missing itself may carry signal.

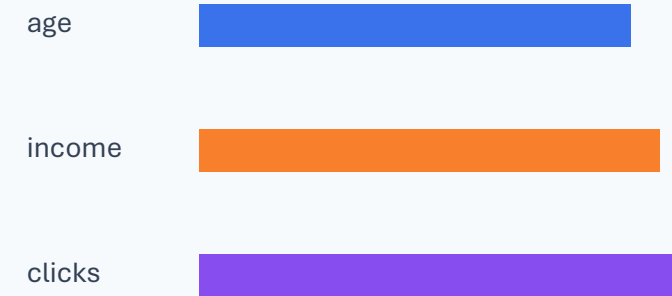


# Scaling makes features comparable

## WITHOUT SCALING



## WITH SCALING



## Algorithms affected most

Distance-based: KNN, clustering, SVM with distance kernels

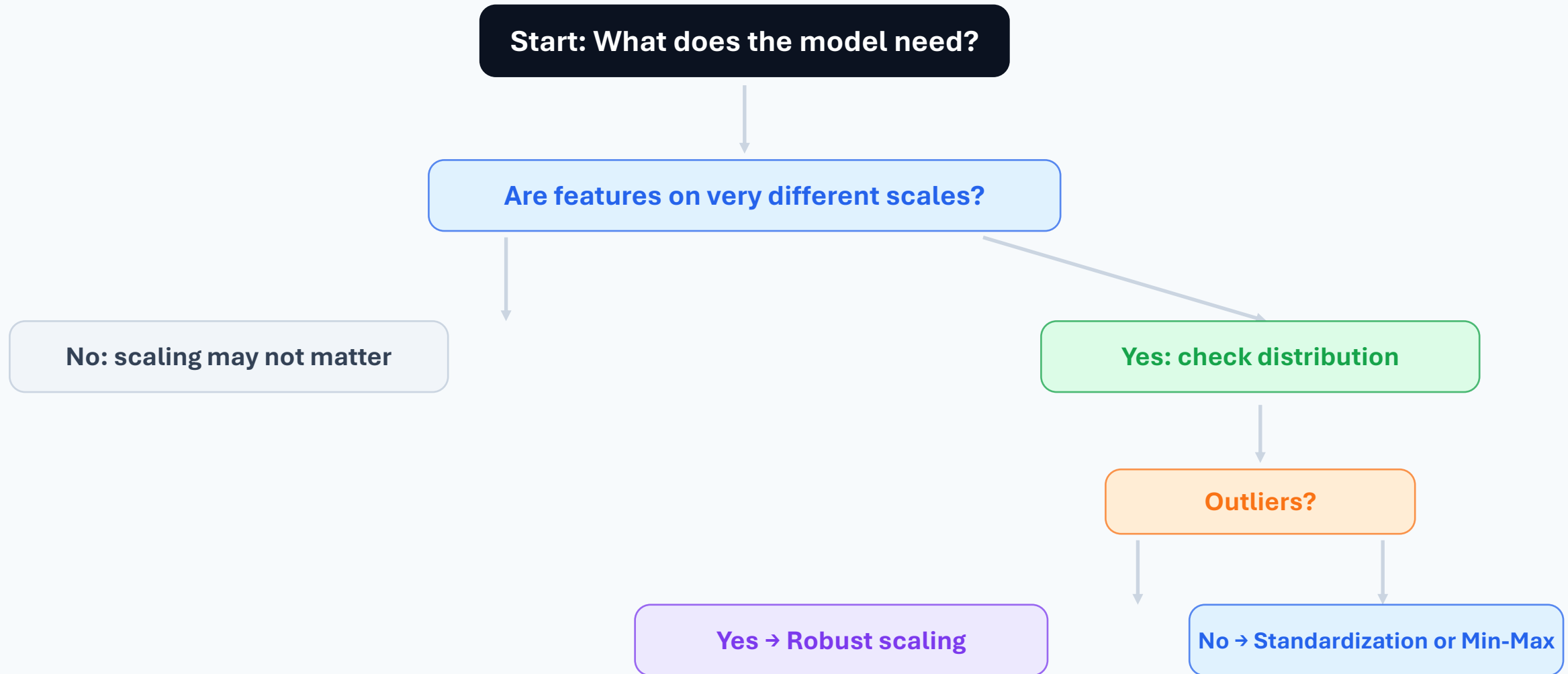
Gradient-based: linear/logistic regression, neural networks

Regularized models: feature scale affects penalties

**Tree-based models usually need less scaling, but pipelines often still standardize for consistency.**



# Choosing a scaler: useful rules of thumb



**Always validate empirically: rules of thumb guide experimentation; they do not replace it.**

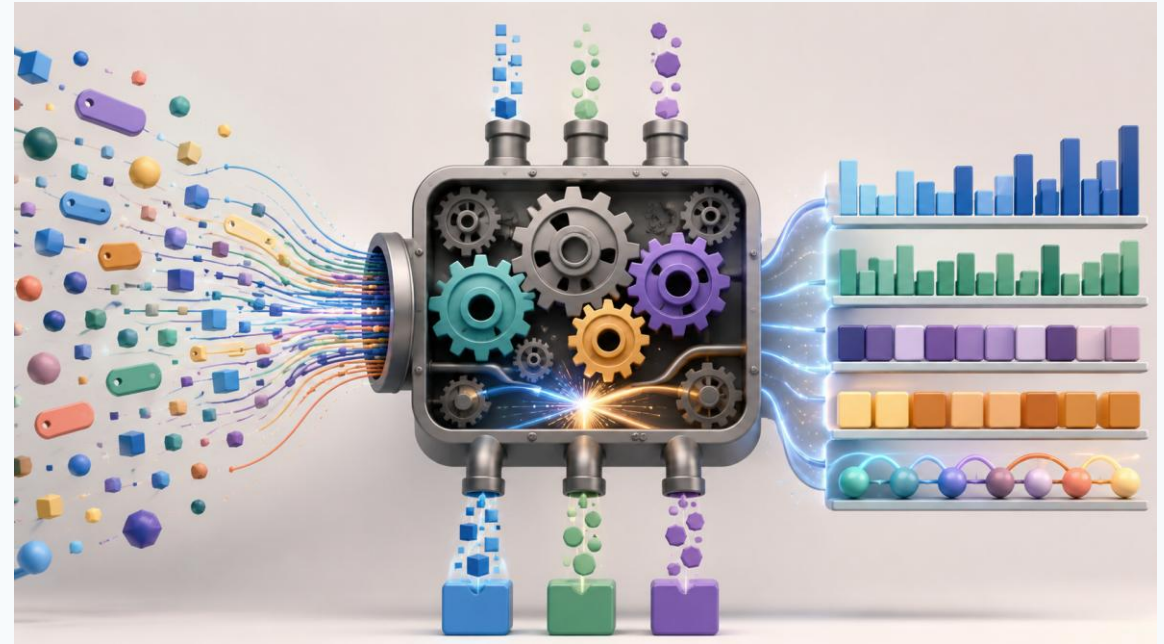
# Feature engineering makes useful signal visible

## Definition

Transform raw columns into features that represent the problem better for predictive models.

**Raw data is not automatically good representation.**

**Good features encode context, relationships, and scale.**



# Encoding categorical variables: turn labels into numbers carefully

## One-hot encoding

Creates one binary column per category. Best when categories are nominal and not too many.

## Label encoding

Maps categories to integers. Use carefully: it can imply false order.

## Target encoding

Replaces a category with a target statistic. Powerful, but leakage-prone.

### EXAMPLE: "PLAN" COLUMN

| plan       | free | pro | enterprise |
|------------|------|-----|------------|
| free       | 1    | 0   | 0          |
| pro        | 0    | 1   | 0          |
| enterprise | 0    | 0   | 1          |

**Watch for high cardinality:  
thousands of categories can explode  
feature space.**

# Creating new features: derived, interaction, and polynomial

## Derived features

Use domain logic to create a more meaningful variable.

**total\_amount = price × quantity**

## Interaction features

Capture a relationship between two variables.

**income × credit\_score**

## Polynomial features

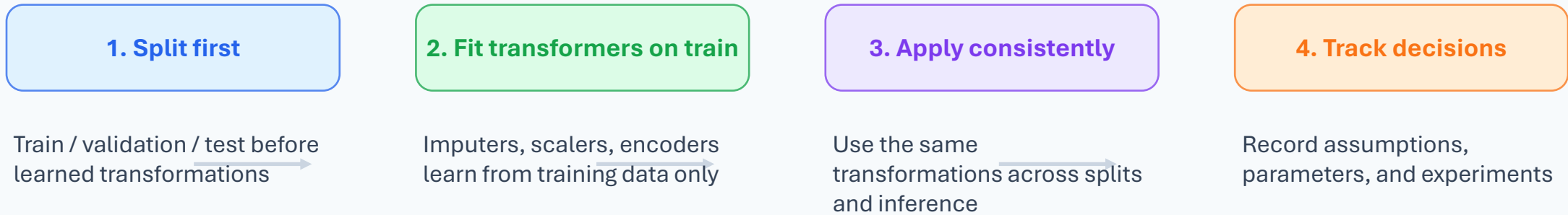
Represent non-linear patterns with powers/combinations.

**$x, x^2, x^3$**

## Better features simplify the model's job.

But every new feature should be justified, tested, and documented.

# A practical preprocessing workflow for ML projects



## SOFTWARE ENGINEERING MINDSET

**A preprocessing pipeline is code that should be reproducible, reviewable, and deployable — not just a temporary notebook experiment.**

# Key takeaways

- ✓ **Clean before modeling**  
Models amplify data quality problems.
- ✓ **Treat missingness as a decision**  
Dropping, imputing, and flagging have trade-offs.
- ✓ **Scale when algorithms care**  
Especially distances, gradients, and regularization.
- ✓ **Engineer features deliberately**  
Make signal visible, but avoid leakage and noise.

